Mining Human–AI Collaboration in Software Engineering Repositories

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***Abstract*—AI coding agents are becoming increasingly involved in software development, but it is still unclear whether human review behaviour changes as repositories gain experience working with them. This study examines review behaviour at the repository level using the AIDev dataset. Our final sample contains 6,147 pull requests from 46 GitHub repositories, including 4,113 agent-authored and 2,034 human-authored pull requests. Each repository's agentic history is divided into early and late periods. We study two measures: formal human review coverage and, among reviewed pull requests, human approval rate. Paired Wilcoxon signed-rank tests are used to compare the early and late periods. We did not find a statistically significant early-to-late change in review coverage for either agent-authored or human-authored pull requests. We also did not find a significant change in approval behaviour. However, agent-authored pull requests had lower median formal review coverage than human-authored pull requests in both periods.**

I. Introduction

AI coding agents are becoming more important in modern software engineering. Unlike traditional code completion tools, newer coding agents can perform larger development tasks, modify several files, and create pull requests that are later reviewed by human developers. The AIDev dataset shows that these tools are already being used in real open-source software projects and are becoming part of normal development workflows [2].

This creates an important question for software teams. Even if an agent is able to produce working code, humans are still responsible for deciding whether the contribution should be accepted. Pull request review is one of the main places where this human oversight can be observed. Developers can inspect the proposed changes, comment on them, request modifications, or approve the pull request.

As developers become more familiar with AI-generated contributions, their review behaviour may also change. Yu et al. studied this problem at the reviewer level and found that individual reviewers became more likely to approve agent-authored pull requests as they gained experience reviewing them, while their commenting effort decreased [1]. They describe this behaviour as habituation.

However, their analysis focuses on reviews that actually happened. A pull request that does not receive a formal review does not create a review observation. This means that another possible change in human oversight is not directly captured: instead of reviewing agent pull requests less carefully, developers might simply review fewer of them over time.

Our study looks at this question from the repository level. Instead of following individual reviewers, we follow repositories through their history of using coding agents. This allows us to measure how often agent-authored pull requests receive formal human reviews and compare them with human-authored pull requests in the same repositories. We also examine whether approval behaviour changes between the early and late stages of each repository's agentic history.

II. Research Questions

A. Review Coverage

How does human review coverage of agent-authored pull requests change from the early to the late stages of a repository's agentic history, compared with human-authored pull requests in the same repositories?

B. Approval Behaviour

Among pull requests that receive human reviews, how does the human approval rate change from the early to the late stages of a repository's agentic history, and does this change differ between agent-authored and human-authored pull requests?

III. Methodology

The main data source used in this study is the AIDev dataset introduced by Li et al. [2]. For reproducibility, the analysis uses the fixed AIDev revision `68ed5f4b`. Using a fixed revision ensures that changes to the upstream dataset do not modify the sample or the results of the study.

The analysis mainly uses three tables from AIDev. The `pull_request.parquet` file contains agent-authored pull requests, `human_pull_request.parquet` contains human-authored pull requests, and `pr_reviews.parquet` contains review information associated with agent-authored pull requests.

For human-authored pull requests, the original review summary did not contain enough information to reliably determine whether every review was submitted by a human or by an automated account. We therefore used a frozen file of detailed review events previously collected from GitHub. This file is included in the replication package, meaning that the final analysis does not depend on live GitHub API calls.

The data were cleaned by checking for duplicated pull requests, missing review information, and repositories without enough data for both PR origins. To support the comparison between agent-authored and human-authored pull requests, repositories were required to contain at least ten pull requests from each origin. One repository had unavailable detailed human review data. It was removed from both the Agentic

and Human groups so that both groups would use the same repository sample.

The final sample contains 46 repositories and 6,147 pull requests. Among these, 4,113 are agent-authored pull requests and 2,034 are human-authored pull requests.

For each repository, its agent-authored pull requests were ordered chronologically. The midpoint of this history was then used to divide the repository into an Early and a Late period. The same time boundary was applied to the human-authored pull requests from that repository. This means that both PR origins are compared using the same definition of Early and Late inside each repository.

For RQ1, review coverage is defined as the proportion of pull requests that received at least one formal human review:

$$\text{Review Coverage} = \frac{\text{PRs with at least one human review}}{\text{Total PRs}}.$$

Review coverage was calculated separately for each repository, PR origin, and period. This results in four groups: Agentic Early, Agentic Late, Human Early, and Human Late.

For RQ2, we examine only pull requests that received formal human reviews. Approval rate is calculated from human review events:

$$\text{Approval Rate} = \frac{\text{Approved human review events}}{\text{Total human review events}}.$$

To avoid calculating approval rates from only one or two review events, the main RQ2 analysis requires at least five human review events for each repository, origin, and period. Fifteen repositories satisfy this condition. We also repeated the analysis with minimum thresholds of three and ten reviews to verify whether changing the threshold affected the overall conclusion.

For both research questions, comparisons were performed at the repository level using two-sided Wilcoxon signed-rank tests. We compare the Early and Late periods for agent-authored pull requests, repeat the same comparison for human-authored pull requests, and then compare the Early-to-Late change between the two origins.

The complete replication package for this study, including the analysis notebooks, cleaned data, generated figures, and documentation, is available in our GitHub repository:

https://github.com/Raselll/LOG6307E-AI-Teammates-Mining

AI tools were used only for grammar, formatting, and language clarity. All data collection, analysis, results, and interpretations were conducted and verified by the authors.

IV. Results

Table I and Fig. 1 present the results for human review coverage. Across the 46 repositories, the median review coverage of agent-authored pull requests was 0.2247 during the Early period and 0.2424 during the Late period. Human-authored pull requests had higher median coverage in both periods, increasing from 0.3594 to 0.5109.

However, the main analysis compares the change inside each repository rather than only comparing the overall period medians. The median per-repository change was -0.0045 for agent-authored pull requests and $+0.0191$ for human-authored pull requests. Neither change was statistically significant. The Wilcoxon test gave $p = 0.388$ for agent-authored pull requests and $p = 0.197$ for human-authored pull requests. The difference between the two Early-to-Late changes was also not significant ($p = 0.167$).

TABLE I
HUMAN REVIEW COVERAGE, EARLY VS. LATE (46 REPOSITORIES)

PR origin	**Early**[a]	**Late**[a]	Δ[b]	p[c]
Agent-authored	0.2247	0.2424	-0.0045	0.388
Human-authored	0.3594	0.5109	$+0.0191$	0.197
Difference[d]	–	–	-0.0828	0.167

[a]Median repository-level review coverage.
[b]Median of the per-repository change (late − early).
[c]Two-sided Wilcoxon signed-rank test.
[d]Agent-authored change minus human-authored change.

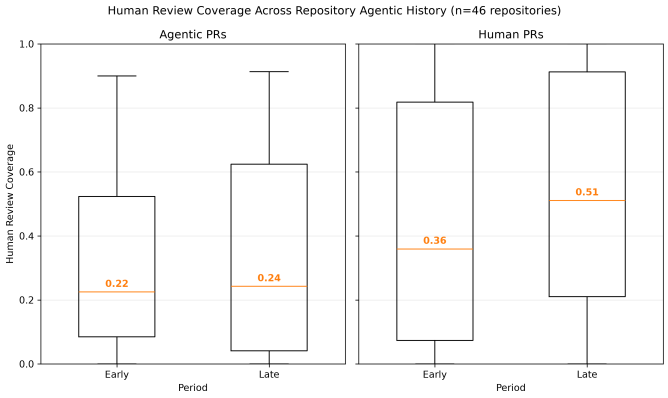


Fig. 1. Human review coverage per repository, early vs. late.

The distributions in Fig. 1 also show important variation between repositories. Some repositories formally review almost none of their pull requests, while others review a much larger proportion. The most visible descriptive difference is therefore not the Early-to-Late change, but the difference in review coverage between agent-authored and human-authored pull requests.

Table II and Fig. 2 present the results for human approval behaviour. After requiring a minimum of five human review events in every repository/origin/period cell, 15 repositories remained in the main RQ2 sample.

The median approval rate of reviewed agent-authored pull requests was 0.5000 in the Early period and 0.3564 in the Late period. For human-authored pull requests, the median approval rate changed from 0.3529 to 0.4012.

Once again, the paired repository-level changes differ from the difference between the two overall medians. The median per-repository change for agent-authored pull requests was $+0.0485$, while the change for human-authored pull requests

was -0.0034. Neither was statistically significant. The corresponding p-values were 0.639 and 0.890. The difference between the changes for agent-authored and human-authored pull requests was also not significant ($p = 0.890$).

TABLE II
HUMAN APPROVAL RATE, EARLY VS. LATE (15 REPOSITORIES)

PR origin	**Early**[a]	**Late**[a]	Δ[b]	p[c]
Agent-authored	0.5000	0.3564	$+0.0485$	0.639
Human-authored	0.3529	0.4012	-0.0034	0.890
Difference[d]	–	–	-0.0118	0.890

[a]Median repository-level approval rate; repositories with at least 5 human reviews per origin and period.
[b]Median of the per-repository change (late − early).
[c]Two-sided Wilcoxon signed-rank test.
[d]Agent-authored change minus human-authored change.

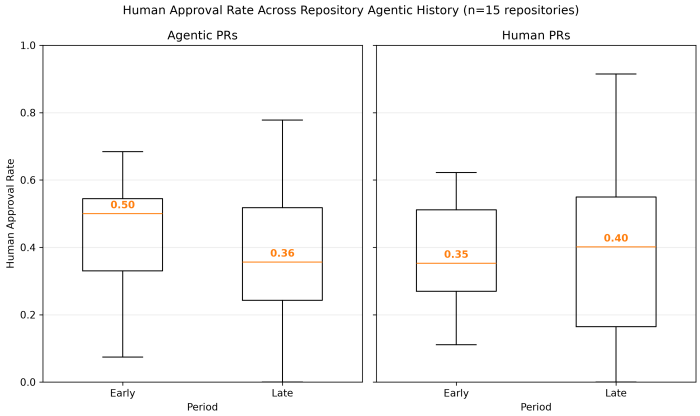


Fig. 2. Human approval rate per repository, early vs. late.

The RQ2 sensitivity analysis gave the same overall conclusion. Using minimum thresholds of three, five, or ten reviews did not produce a statistically significant change in approval rate for either PR origin, and the difference between their trajectories also remained non-significant.

V. DISCUSSION

Based on the 46 repositories analyzed, human review coverage did not show any statistically significant change from the early to the late stage of the repositories' agentic histories. The p-values for agent-authored PRs (p = 0.39), human-authored PRs (p = 0.20), and the difference between the two (p = 0.17) are all high. Approval of reviewed agent PRs, measured over 15 repositories, also showed no significant change (p = 0.64), and neither did approval of human PRs nor the difference between the two (both p = 0.89). The small sample size and the large variation between the repositories studied limit what we could have detected, so these results are compatible with either no change or a change too small to see. However, the typical repository's agent coverage barely moved, so there is no sign of a large shift.

Our results do not contradict those of Yu et al., who followed 400 reviewers through their reviewing experience and found that their approval of agent PRs rose over time, whereas we followed repositories through their agentic histories. Our approval results, which show no significant rise, have two possible explanations. Reviewers might become more lenient with experience, but new reviewers joining over the repository's life start out stricter, which could counteract this behavior and keep the approval rate flat. Unfortunately, this hypothesis could not be tested, as the sample had too few reviewers per period. The second is limited power: with only 15 repositories, a change could be harder to detect. Coverage, on the other hand, has no counterpart in their study.

The most visible pattern in the results is not the change from early to late, but rather the difference in coverage level between agent-authored and human-authored PRs. In both periods, the median review coverage of agent PRs (22% and 24%) is lower than that of human PRs (36% and 51%), but no test checked whether this difference is real. Moreover, formal reviews are only a lower bound on human oversight, since PRs can also be examined informally, or merged or closed without a review. Because each repository is compared with itself, an undercount that stays constant does not distort the early-versus-late comparison.

For teams working with coding agents, this suggests that checking review coverage might not be sufficient to evaluate the oversight that agent-authored PRs receive. Since we found no evidence that coverage rises or falls on its own, teams should not count on it improving with time. Either including a broader range of metrics from existing data, or putting a review policy in place, could help these teams make human oversight of their agents' PRs visible. Finally, Yu et al. recommend rotating reviewers to keep individual approval from drifting upward; our turnover hypothesis is consistent with this recommendation, though we did not test it.

Another practical point is that the absence of a significant temporal change should not be interpreted as evidence that review practices are already sufficient. Our analysis measures whether review behaviour changes over time, not whether the observed level of review is appropriate. The lower descriptive review coverage of agent-authored PRs suggests that teams may still benefit from explicitly monitoring how agent contributions are reviewed. In practice, repositories could track review coverage together with approval rate, requested changes, and review comments rather than relying on a single measure. This would provide a broader view of human oversight and could help identify changes that are not visible through formal approval data alone.

VI. THREATS TO VALIDITY

Construct validity. Our main metric, formal GitHub PR reviews, is only a part of a PR's control. PRs can be examined and then merged or closed without a review, discussed in comments, or discussed outside of the PR. This could distort the results if humans or agents have a larger share of their PRs merged or closed without a review. Human-authored PRs may also contain AI-assisted code, which is close to impossible to evaluate but blurs the line between the two groups, so a real

difference between them would be understated. Finally, agents do not all behave the same way in terms of PR authoring. Devin and Copilot open their PRs from a bot account, whereas Codex, Cursor and Claude Code require a human to open the PR. Since the author of a PR cannot approve it, the latter require two humans to issue and approve a PR, making formal approval harder to obtain. "Agent-authored" is therefore not one homogeneous group.

Internal validity. The early and late periods are defined by the midpoint of the agent-authored PR count. These periods are then applied to the human-authored PRs, meaning that their length and density can differ. A two-period system can also show only net differences, meaning that a bell-shaped pattern of agent usage over time would be identified as stable.

Conclusion validity. With 46 repositories, our RQ1 tests have a larger sample than RQ2, but a small change could still exist without being identified. With only 15 repositories for RQ2, even a meaningful change could be missed. Low power favours finding nothing, which is the result we got. Comparing every repository with itself removes differences between repositories, but does not compensate for the small sample.

External validity. Our findings apply to repositories that are popular (more than 100 stars), public and maintained on GitHub, receiving a steady PR flow from both humans and agents. In our sample, agent PRs are reviewed more often than in the full dataset (about 36% against 24%), and most agent PRs come from Devin (57% against 14%). Our observations may therefore not transfer to private projects, small or single-developer projects, Codex-dominated repositories, or later stages of adoption.

VII. Conclusion

This study examined whether human review behaviour changes as repositories gain experience working with AI coding agents. Across the 46 repositories in the RQ1 sample, we did not find a statistically significant Early-to-Late change in formal review coverage for either agent-authored or human-authored pull requests. We also did not find a significant difference between their changes.

For approval behaviour, the main analysis included 15 repositories with enough review activity. Again, we did not find a statistically significant change for agent-authored pull requests, human-authored pull requests, or the difference between the two groups.

These results do not mean that human review behaviour never changes when teams adopt coding agents. Formal review captures only one part of human oversight, and the smaller sample used for approval behaviour limits the conclusions that can be drawn. However, within the repositories and period studied here, we did not observe evidence of a large repository-level shift in either review coverage or approval behaviour.

Future work could include other forms of oversight such as comments, review duration, requested changes, merge behaviour, or informal review activity. It could also combine repository-level and reviewer-level analysis to better understand whether individual reviewer habituation eventually produces changes that can be observed at the repository level.

References

[1] H. Yu, L. Liu, X. Jiang, Y. Jia, S. Wang, P. Qian, and Y. Chen, "Habituation at the gate: Rising approval and declining scrutiny in human review of AI agent code," arXiv preprint arXiv:2606.22721, Jun. 2026.

[2] H. Li, H. Zhang, and A. E. Hassan, "The rise of AI teammates in software engineering (SE) 3.0: How autonomous coding agents are reshaping software engineering," arXiv preprint arXiv:2507.15003, 2025.